



# Coding a Disability Inclusive World

**3 Stories**

2024 // Dennie Declercq //

Dutch Umbraco Experience

# Dennie Declercq

- Dennie Declercq
- President/ IT-Coach DDSoft
- Microsoft MVP
- Mr. #DreamingIsBelieving
- <https://ware.ddsoft.be>





# Content

- Intro: 2030
- Stories
- Persona's
- Integration

# 2030

---

- Imagine...
- When we continue doing what we do
- Digitalization
- Without a huge accessibility implementation revamp
- Focus: Software for 'end-users'/ citizens
- Ex.: Banking apps, Mobile wallets, Online ticketing, Online shopping, ...





Down  
Syndrome

\$ \$ \$



John

# John's Profile

## Disability

Down Syndrome

---

A Learning  
Disability

## Needs

Iconography & Pictures

---

Speech

---

Easy Language

---

Low Budget Devices

---

In Person Interactions

# Down Syndrome: A learning Disability

## Intelligence Functioning

Adults: <75

---

Illiteracy

---

Difficulties with Math

## Adaptive Behavior

Difficulties with envisioning

---

Missing reasoning

---

Different imagination

---

Problems daily life

---

Communication



## Easy Language

- Avoid Jargon all time
- Short sentences.
- Easy word variations
  - (ex: a ticket vs your accreditation)
- Avoid Acronyms

Show your ticket at the entrance.  
Ring the bell to the left when nobody is here.  
Enter your date of birth here:  
Type your name here:



## Iconography & Pictures

- What is: 烤雞?





# Speech

- Add speech interfaces
- Cloud or mobile
- Azure Speech
- Device Text-To-Speech



**Azure AI services | Speech service**  
Azure AI services



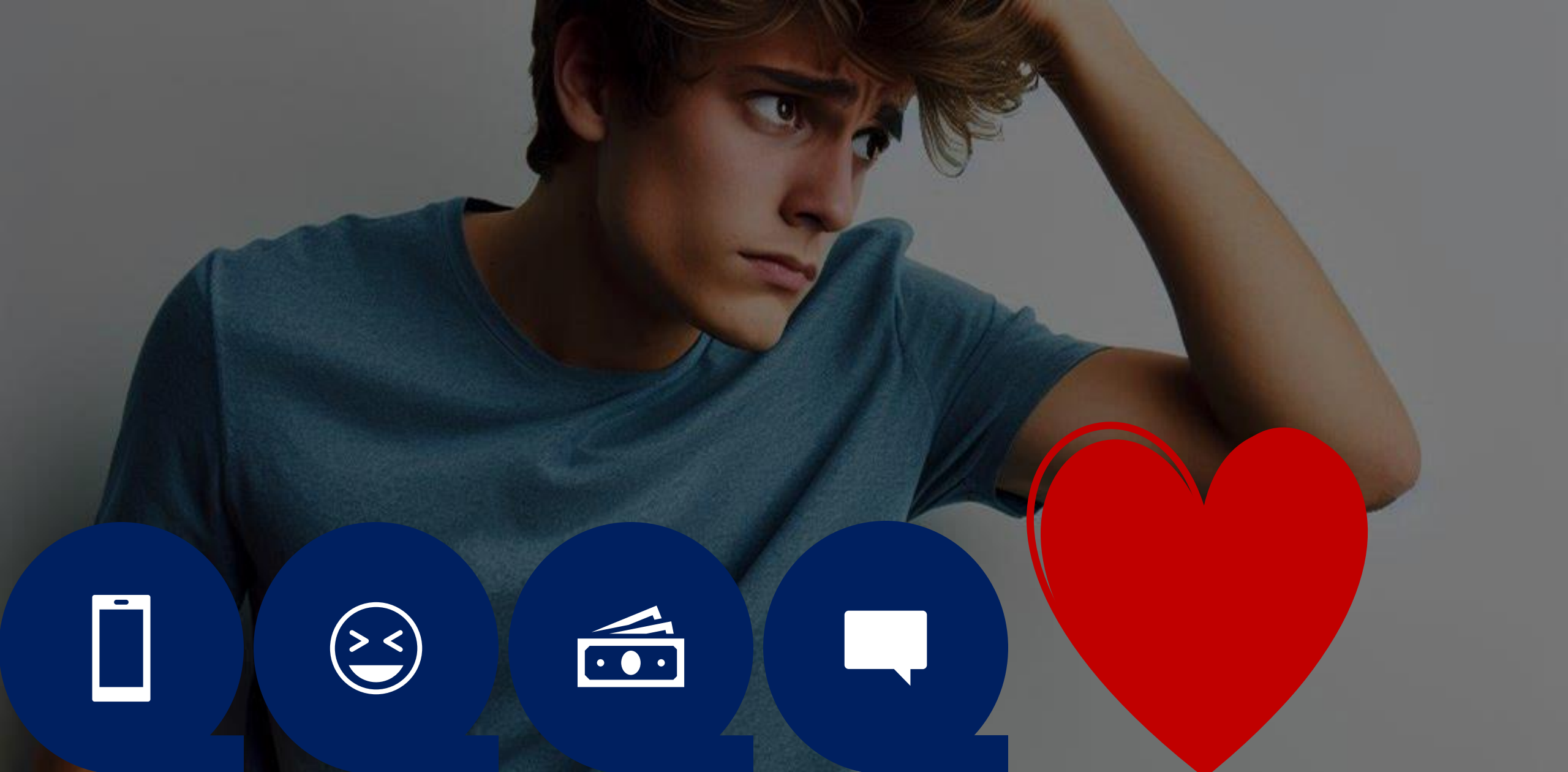
## Low Budget Devices

- PWD often have lesser income
- Not always the latest (& safest) device
- Publish software:
  - Tested on low-end devices
  - Tested on older Android/ iPhone/ Windows



## In Person Interactions

- I know, we are developers & geeks. But...
- Not everything need to be digital
- In specific: Not Only Digital
- In person interactions are sometimes needed
- Human In the loop
- Human Interception
- Digital Twin of Physical Action



2030: Try Again

John

Autism  
(Teenager)



Lisa

# Lisa's Profile

## Disability

Autism  
(child)

## Needs

More structure

---

More Time

---

More Exercises

---

Trust

# Autism: A Disability or Talent?

## Disability

Rigidity

---

Difficulties with structure

---

Difficulties with communication

---

‘Own World’

---

Challenges with stress

## Talent

Deep Focus

---

Special Interest

---

Out-of-the-Box

# Children on the spectrum

## Children

Structure missing

---

‘Different behavior’

---

Shutdown

---

Meltdown

---

Reasoning/ Rigidity ➔ Seen as ‘stubborn’

## Adults

Need help with structure

---

Help with communication

---

Often find for most stuff ‘work-arounds’

---

Or... Coping...



## Structure

- Apps, Software & Sites providing structure
- Clear and easy to find opening hours, street address, contact info..
- Straightforward & up to date information



## Time

- Provide visualization of time
- Provide option to extend time



## Exercises (Gamification)

- Gamification for learning/ study apps & websites
- Gamification for mental health & activity tracking software
- Use dopamine feedback system



## Building Trust

- Limit software crashes & errors
- Use settings easy to change
- Right & easy usage of parental control



2030: Try Again

Lisa



Brain Injury

Dementia



Kim & Dave

# Kim 's profile

## Disability

Acquired  
Brain Injury

## Needs

Pictures

---

Speech

---

Simple Layout

---

Human Interaction

# Dave 's profile

Disability

Dementia

Needs

Loud Devices

---

Physical Assets

---

Memories

---

Human

# Disability: Acquired Brain Injury

## Disability

Intelligence decrease

---

Concentration decrease

---

Due to Accident or  
disease

## Needs

Simple layout

---

Human-in-the-loop

---

Easy Language

# Disability: Dementia

## Disability

Decreased Brain Functioning

---

Decreased memory

---

Not able to learn new things

---

Staying in the past

## Needs

Reminiscence

---

Loud Devices

---

Physical Assets



## Simple Layouts

- Bright backgrounds
- Large fonts
- Limited need for navigation
- Icons



## Loud Devices

- Limit usage of “Maximum Volume” blocker
- Older or specialized hardware



## Reminiscence

- Use questionnaires about past life
- Images and quotes like: “On this day in ....”
- Simulate family interaction



# Human

- Human take-over (telephone centers ex: Hospital contact center)
- Human the loop (talking to a person)
- Human interception (make exceptions in online forms)



## Physical Assets

- Real money
- Real physical pictures
- Old parking system with buttons
- Easy tickets at public transport, meeting center,...



Kim & Dave - Coming Soon ..



Persona's

# What are persona's

- A possible person
- A recognizable name
- Basic Description of the person
- Positive wording



# Persona mapping (Example Hearing Herald)



# Personas: John, Lisa, Kim & Dave

| Recognizable Name  | Basic Personal Information                      | Topics of Interests                      | Disabilities & Difficulties           |
|--------------------|-------------------------------------------------|------------------------------------------|---------------------------------------|
| Learning Larry     | 20 Yo;<br>Male;<br>Volunteer;<br>Single         | Pub;<br>His friends;<br>His Grandpa      | Down Syndrome;<br>Learning Disability |
| Authentic Ailsa    | 15 Yo;<br>Female;<br>Student;<br>Single         | Metal Music;<br>Girlfriend;<br>Going out | Autism (Teen/Child)                   |
| Brainy Brené       | 56 Yo;<br>Female;<br>Sickness leave;<br>Married | Family;<br>Nature                        | Acquired Brain Injury                 |
| Remembering Ronald | 66 Yo;<br>Male;<br>Retired;<br>Married          | Restaurants;<br>Snooker                  | Dementia                              |

# User Research

- Look at your current users
- Look at potential users
- Ask about their difficulties

# Integration

---

Using's Persona's needs in  
the Software Roadmap



# Linking Persona's Needs

- Lists Persona's Disabilities/Difficulties
- Accessible Accommodations an Approach for each disability/difficulty
- Nice to have for each Persona

## Linking Needs (ex: Hearing Herald)

| Disability         | Needs                         | Nice-to-Have                  |
|--------------------|-------------------------------|-------------------------------|
| Hard of Hearing    | Loud device output            | Assistive Hearing aid support |
|                    | Subtitles                     |                               |
| Not digital minded | Easy Lay-out                  | Exercise interface            |
|                    | Bug-free stability            |                               |
|                    | Human interception/ take-over |                               |

# Needs: John & Lisa

| Disability               | Needs                         | Nice-to-Have          |
|--------------------------|-------------------------------|-----------------------|
| Persona: Learning Larry  |                               |                       |
| Down Syndrome            | Iconography                   | Pictures              |
|                          | Easy Language                 |                       |
|                          | Low Budget device Interop.    | Text-To-Speech        |
|                          | Human Interception/ take-over |                       |
|                          |                               |                       |
| Persona: Authentic Ailsa |                               |                       |
| Autism (Kids/ Teens)     | Structure                     | Gamifications         |
|                          | Time                          |                       |
|                          | Exercises                     | Easy Parental Control |
|                          | Trust mechanisms              |                       |

## Needs: Kim & Dave

| Disability                  | Needs             | Nice-to-Have   |
|-----------------------------|-------------------|----------------|
| Persona: Brainy Brené       |                   |                |
| Acquired Brain Injury       | Pictures          | Text-To-Speech |
|                             | Human-in-the-loop |                |
|                             | Easy Language     |                |
|                             |                   |                |
|                             |                   |                |
| Persona: Remembering Ronald |                   |                |
| Dementia                    | Loud Devices      | Reminiscence   |
|                             | Human             |                |
|                             | Physical Assets   |                |
|                             |                   |                |

2030: Try Again.  
What About Kim & Dave?



2030: Try Again

Kim & Dave



## Q&A || Thanks

- [Dennie@DDSoft.be](mailto:Dennie@DDSoft.be)
- @DennieDeclercq
- #DreamingIsBelieving
- <https://ware.ddsoft.be>



Review my session